Windows CMD → Linux CLI Command Reference

Navigation & Files

Task	Windows CMD	Linux CLI
Show current directory	`cd`	`pwd`
Change directory	`cd foldername`	`cd foldername`
Go up one level	`cd ..`	`cd ..`
Go to root	`cd \`	`cd /`
Go to home	`cd %USERPROFILE%`	`cd ~`
List files	`dir`	`ls`
List with details	`dir /a`	`ls -la`
Clear screen	`cls`	`clear`

File Operations

Task	Windows CMD	Linux CLI
Create file	`type nul > file.txt`	`touch file.txt`
Create folder	`mkdir folder`	`mkdir folder`
Copy file	`copy file.txt dest\`	`cp file.txt dest/`
Move file	`move file.txt dest\`	`mv file.txt dest/`
Rename file	`rename old.txt new.txt`	`mv old.txt new.txt`
Delete file	`del file.txt`	`rm file.txt`
Delete folder	`rmdir /s folder`	`rm -r folder`
Print file contents	`type file.txt`	`cat file.txt`

Search & Find

Task	Windows CMD	Linux CLI
Find text in file	`findstr "word" file.txt`	`grep "word" file.txt`
Find files	`dir /s filename`	`find . -name "filename"`
Search recursively	`findstr /s "word" *.*`	`grep -r "word" .`

System Info

Task	Windows CMD	Linux CLI
Show IP address	`ipconfig`	`ip a` or `ifconfig`
Ping	`ping google.com`	`ping google.com`
Running processes	`tasklist`	`ps aux`
Kill a process	`taskkill /PID 1234`	`kill 1234`
Disk usage	`diskpart` / `wmic`	`df -h`
Folder size	—	`du -sh folder/`
Who am I	`whoami`	`whoami`
Environment vars	`set`	`env` or `printenv`
Single env var	`echo %PATH%`	`echo $PATH`

Network

Task	Windows CMD	Linux CLI
Trace route	`tracert google.com`	`traceroute google.com`
DNS lookup	`nslookup domain.com`	`nslookup domain.com` or `dig domain.com`
Open ports	`netstat -an`	`netstat -tulnp` or `ss -tulnp`

Package / Software (Linux only concept)

Linux has a package manager — Windows CMD has no direct equivalent:

```
sudo apt update          # Refresh package list (Debian/Ubuntu)
sudo apt install curl    # Install a package
sudo apt remove curl     # Remove a package
sudo yum install curl    # On RHEL/CentOS
```

Key Differences to Remember

- **Paths:** Windows uses `\`, Linux uses `/`
- **Drive letters:** Windows has `C:\`, Linux has one root `/`
- **Case sensitivity:** Linux filenames ARE case-sensitive (`File.txt ≠ file.txt`)
- **Permissions:** Linux uses `chmod 755 file` / `chown user file` — no CMD equivalent
- **Superuser:** Windows uses `Run as Administrator`; Linux uses `sudo` before a command
- **Line endings:** Windows files use `\r\n`; Linux uses `\n` (matters when scripting)
- **Wildcards:** Both use `*`, but Linux also uses `?` for single char and `[abc]` patterns more broadly

Helpful Linux-only Shortcuts

```
!!          # Repeat last command
!ls         # Repeat last command starting with 'ls'
Ctrl+C      # Cancel running command
Ctrl+Z      # Suspend process
Ctrl+R      # Search command history
history     # Show command history
man ls      # Manual/help for any command (like /? in CMD)
ls --help   # Shorter help
```